ASHISH AGGRAWAL

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Systems & Machine Learning Engineer transitioning from 10+ years of global operational leadership—most recently at Amazon—to low-level systems programming. Expert in bridging the gap between hardware optimization (C++ / Linux Kernel) and predictive intelligence (AI / ML). Leveraging a decade of high-stakes problem-solving to architect automated, scalable architectures for High-Frequency Trading (HFT) and OS Kernel environments.

TECHNICAL SKILLS

- **Languages**: C++, C, Python, Bash, x86 Assembly
- **Systems**: Multithreading, IPC, Kernel Development, Device Drivers, Memory Management
- **ML/AI**: Neural Networks, NLP, Computer Vision (OpenCV), LLM Integration, Statistical Modeling
- **Tools**: Linux, Git, Docker, Make/CMake, Vim

PROJECTS

1. Safety-Critical Multithreaded Watchdog | C++

- **Built a real-time monitoring system** designed to trigger an immediate **Emergency Stop (E-Stop)** if the primary control software hangs or misses a **strict 500ms heartbeat**.
- Prevented **deadlocks and priority inversion** by using a lock-free communication path using C++ **atomics**, ensuring the monitor thread stays independent of the main process.
- **Stress-tested** for deterministic failure by developing a fault-injection suite that simulated CPU starvation and thread crashes to guarantee the system fails safely under pressure.
- **Tech:** C++20, POSIX Threads (pthread), Atomics, Linux (SCHED_FIFO).

2. Lock-Free Concurrent Queue | C

- **Developed a high-throughput, non-blocking queue** using the Michael & Scott algorithm to remove the performance bottlenecks typically caused by mutex contention.
- **Ensured thread-safety across 32+ concurrent threads** by implementing atomic primitives and compare_exchange_strong for linearizable operations.
- **Solved complex memory management issues** by implementing a deferred reclamation strategy, validating the entire system through a rigorous suite of 190+ automated test cases.
- **Tech**: C, Atomics, Memory Barriers, Concurrency Testing.

3. Multi-Type Financial Fraud Detector| Python

- Created a **smart fraud detection system** that automatically recognizes data types to spot **credit card fraud, tax evasion, and invoice errors** across different financial sectors.
- Used Machine Learning (**Isolation Forest**) and custom rules to flag suspicious patterns and assign a **0–100 risk score**, helping auditors focus on the most critical threats first.
- Developed an interactive dashboard using **Streamlit and Plotly** to **show real-time fraud trends** and export reports for easy compliance and investigation.
- **Tech Stack:** Python, Scikit-learn, Streamlit, Pandas, Plotly, NumPy.

PROFESSIONAL EXPERIENCE

Amazon Air | Senior Program Manager *Ontario, Canada | Aug 2022 – Sept 2023*

- **ML & Automation**: Directed the implementation of internal **ML-based automation** models to optimize volume capacity planning across North American sites by partnering with Data Scientists and Software Engineers to train predictive algorithms on historical load data.
- **Data Engineering**: Engineered capacity analysis frameworks to resolve critical throughput bottlenecks by utilizing SQL queries to deep-dive massive Amazon Air datasets and collaborating with BIEs to visualize utilization gaps.
- Overhauled operational planning protocols to accelerate the adoption of new research capabilities by executing internal "**Go-to-Market**" strategies and **change management** frameworks for tool deployment.

Amazon Logistics | Site Operations Manager *Ontario, Canada | Aug 2020 – July 2022*

Managed high-volume operations for a 500,000 sq. ft. facility, leveraging analytical tools to drive a 22% productivity increase and over $2.5M in projected cost savings. Architected network-level logistics programs and orchestrate critical load planning, achieving region-leading throughput (51.2 TPH) through dynamic resource allocation and system stability modeling**.**

Prior Operational Management Experience

Includes roles at ***Fresh Start Foods, Maxxam Analytics,*** *and* ***Vikram Laboratories*** Successfully handled operations across manufacturing and lab environments with a focus on process efficiency and fault tolerance. Refactored scheduling logic to eliminate bottlenecks, increasing output by 77% and uptime by 90%. Enforced rigorous QC standards to maintain zero data integrity failures in high-volume workflows and re-engineered procedures to cut setup times by 20%.

EDUCATION

Bachelor of Science (Honours), Computer Science Trent University, Peterborough, ON | Expected April 2027

- *Distinction:* Dean's Honour Roll.
- *Key Coursework:* Operating Systems, Systems Programming, Computer Networks, Computer Organization, LLMs & NLP, Applied AI, Data Mining, Probability 1, Linear Algebra, Calculus I/II.

Bachelor of Technology, Biotechnology Rai Foundation Engineering College, India